

Motor Imagery based Multivariate EEG Signal Classification for Brain Controlled Interface Applications

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Abstract— Brain computer interface (BCI) can be defined as a pathway that enables human brain to communicate and voluntarily command an external device and generate output instead of depending upon peripheral nerves and muscular movements. Achieving maximum classification accuracy is the greatest challenge in developing a BCI system to correctly interpret the brain signals. This paper aims at investigating various classification algorithms in combination with different pre-processing techniques and comparing their results for maximum classification accuracy. Independent component analysis (ICA), principal component analysis (PCA) and notch filters are used for artifact removal, dimension reduction and noise cancellation, respectively. Left and right hand movements were recorded from the scalp using non-invasive electrodes. Fine KNN, with independent components as feature, gives highest classification accuracy in comparison with various classification techniques used in this research.

Key words: Brain Computer Interface, Electroencephalogram, EEG classification

I. INTRODUCTION

All around the world people suffer from diseases that affect their muscles or cause limb amputations. People suffering from neuromuscular diseases like amyotrophic lateral sclerosis, brainstem stroke or muscular dystrophy still have an intact Central nervous system (CNS) [1]. Brain computer interface (BCI) is a technology that scales the activity of CNS and allows voluntary commands to be translated into artificial outputs such as controlling cursors or a prosthetic limb thus providing a channel of communication between the two [2,3]. This communication between human brain and electronic devices is particularly helpful for patients who are suffering from such disabilities as it makes them unable to generate a motor response to environmental stimuli. Fig.1 shows the steps for classification of EEG data

which includes data acquisition, pre-processing and feature extraction.

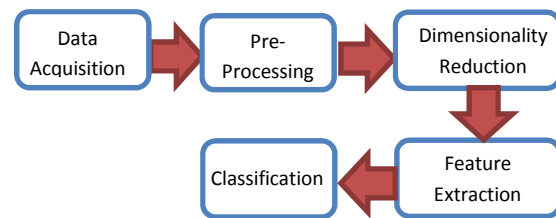


Fig. 1 EEG signal classification process

Brain signals are acquired for further processing and correct classification of intended movement with maximum accuracy. For acquiring EEG data, both invasive and non-invasive methods are used in literature though non-invasive techniques are more popular since they do not require implants [4]. Emotiv EPOC Headset is used for data acquisition in this research because of ease of use [5]. For the purpose of pre-processing, Butterworth notch filter is applied to obtain a band of 14-30 Hz. This is to preserve the required frequency band i.e. beta band [6]. ICA is applied to remove biological artifacts from data [7,8] and PCA is applied to reduce the dimensions of data [9]. Common spatial pattern (CSP) is widely used as a feature extraction technique [10]. While, classifiers that can be applied on the extracted features are linear discriminant analysis (LDA), quadratic discriminant analysis (QDA), Support vector machine (SVM), Decision Trees and K Nearest Neighbor (KNN) algorithms.

Previously researchers have attempted to classify data using different techniques. S. Bhattacharyya et al. presented performance analysis of LDA, QDA, KNN, SVM, and naïve Bayesian classifiers for left and right hand motor imagery

[11,12]. K Tavakolian et al. classified two class motor imagery using adaptive auto regressive coefficient feature with Bayesian classifier, neural network (NN) and Bayes quadratic classifiers to achieve 16.43%, 15.71% and 17.14% classification accuracies, respectively [13]. A novel approach is presented in [14], that combines feature selection with SVM emotion classifiers. In [15], a supervised filter method is proposed for multi-objective feature selection for EEG classification. A new sparse common spatial pattern algorithm is proposed by Arvaneh et. al. in [16] for EEG chaneel selection for BCI. D. Planelles et al. have used SVM to achieve 72 % accuracy to detect arm movement intent [17].

The aim of this research is to evaluate and compare accuracies of different EEG signal classification algorithms with combinations of feature extraction and pre-processing techniques. This will enable the users to choose the most appropriate combination based on the properties of these techniques and their reliability.

Rest of the paper is organized as followed: section 2 describes data acquisition approach, section 3 describes the pre-processing techniques used and section 4 describes dimensionality reduction method used. Feature extraction and classification techniques are explained in section 5 and 6 respectively. Comparison of these techniques is presented in section 7 moreover, section 8 concludes the paper.

II. DATA ACQUISITION

Motor imagery is defined as a state in which subject intends to perform a certain action or mentally simulates it. Data acquisition of EEG signals is mostly based on motor imagery. Data acquisition required for this research was done using 14 channel Emotiv Headset. It consists of two reference electrodes. Electrodes are located using the universal 10-20 electrode placement system according to which the distances between adjacent electrodes is 10-20 percent of the total distance between Nasion and Inion i.e. front and back of the scalp[18]. Electrodes are labeled according to the part of brain over which they are present like: pre-frontal as Fp, frontal as F, temporal as T, parietal as P, occipital as O, and central as C [19]. Meanwhile even numbered electrodes cover right side of the brain while odd numbered electrodes cover left side of the brain.

The paradigm, as shown in Fig. 2 consists of 4 second of fixation cross display followed by 2 seconds cue display and motor imagery for 3 seconds. Motor imagery is the imaginative movement, whereas feedback is recording of signal for motor imagery or rest duration. Cues are displayed as arrow pointers in up and down directions to record two class data. Data for rest period is logged when fixation cross is displayed.

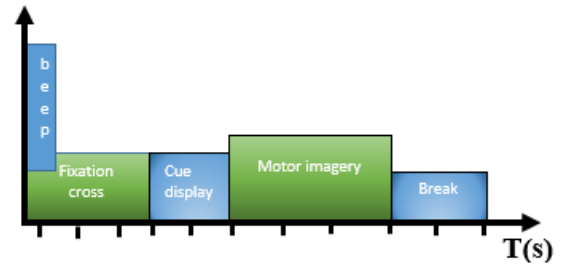


Fig. 2 Paradigm of data recording

The data is recorded for 4 healthy subjects 3 males and 1 female who volunteered for this research. The data consist of consists of 4 trials of each movement. Sampling frequency is kept at 128 Hz. Open Vibe is used as a data collection software which displays the amplitude change of microvolts in tangible signal form and records the signals in a .CSV format which is MATLAB compatible.

III. PRE-PROCESSING

The signal acquired from brain consists of divisions based on frequency band variations (depending on rhythmic oscillations) and transient response. EEG signal is divided into six frequency band categories: Alpha (8-13 Hz), Beta (14-30 Hz), Gamma (36-44 Hz), Theta (4-7 Hz), and Delta (< 4 Hz). First step towards EEG classification is filtering and extracting certain bands of frequency which exhibit the maximum indication of motor imagery. EEG data is filtered in beta band which is the excited state frequency band. For that purpose a tenth order Butterworth notch filter is used.

IV. DIMENSIONALITY REDUCTION

Principal component analysis (PCA) is a dimensionality reduction technique. PCA, while keeping hold of patterns and trends, simplifies the complications in high-dimensional data. It achieves this goal by transforming the data into lesser dimensions, while conserving the maximum information. PCA describes the data with number of orthogonal directions by finding the least error. In PCA, these directions do not reveal the class of the data. PCA is utilized for segmentation of EEG signals from multiple channels. It will be immensely helpful if the number of independent components is known [20]. The basic approach for finding principal components is:

Step 1: Standardization

$$z = \frac{\text{value} - \text{mean}}{\text{standard deviation}} \quad (1)$$

Step 2: Covariance Matrix computation

$$\begin{bmatrix} C(x, x) & C(x, y) & C(x, z) \\ C(y, x) & C(y, y) & C(y, z) \\ C(z, x) & C(z, y) & C(z, z) \end{bmatrix} \quad (2)$$

Where C is a function of Covariance.

Step 3: Principal components are identified by computing the eigenvalues and eigenvectors of the above calculated matrix

Step 4: Feature vector formulation

Step 5: Recast the data along the principal components axes

$$\begin{aligned} & \text{FinalDataSet} \\ &= \text{FeatureVector}^T \\ & * \text{StandardizedOriginalDataSet}^T \end{aligned} \quad (3)$$

Dimension reduction is often suggested before applying classification algorithms to reduce the amount of data to be processed thus saving time and computational power.

V. FEATURE EXTRACTION

After obtaining the required frequency band, and reducing dimensions of the data, ICA is applied on the data for feature extraction. ICA is used to decompose a signal into additive components: The subcomponents are non-Gaussian signals that are considered to be uncorrelated to each other. Without enough information about mixing process and the source signal, ICA is used to separate the signal from mixed signals [21]. Independent component weights W are extracted from the source signal, which are then multiplied with uncorrelated data to evaluate mixing matrix using (4). The mixing matrix M is then subtracted from the original signal to get what we call 'independent components' of the signal as denoted by (5).

$$M = W * \text{source signal} \quad (4)$$

$$\text{EEG}_{\text{pure}} = \text{source signal} - M \quad (5)$$

It has been practically proven that ICA results are best if independent source signals are used and the source signal have non-normal distribution [22]. We approach the determination of source signals by maximum-likelihood techniques. We use an estimate of the density and seek the parameter vector 'a' that minimizes the difference between the source distribution and the estimate. That is, 'a' is the basis vectors of source signal and thus is an estimate of the M [23].

VI. CLASSIFICATION

A. Discriminant Analysis

Discriminant Analysis (DA) is normally used in Artificial intelligence and machine learning for finding combinations between features depicting a number of classes or categories. This combination of results can be used as a classifier between two classes, or as a dimension reduction tool in case of larger number of classes. It addresses whether a feature or feature set is best source of categorizing between two classes. DA works when the measurements are continuous such as in the case of an EEG signal. This algorithm can be considered as "supervised learning" and computes the directions ("discriminants") that will represent the axes with maximum separation between multiple classes [24]. Following are five summarized steps of how DA works:

1. The mean vector for each class of the data set are computed.
2. Scatter matrices are computed.

3. From the scatter matrices, eigenvectors and eigenvalues are computed.
4. A $d \times m$ dimension W matrix is formed by sorting the eigenvectors in decreasing order of their eigenvalues and choosing the m largest eigenvalue eigenvectors.
5. The samples are transformed onto the new subspace by using the W matrix. This can be done by multiplying a $p \times d$ dimensional matrix X by the W matrix to get $p \times m$ dimensional Y matrix.

Two types of discriminant analysis that can be used are linear discriminant analysis (LDA) and quadratic discriminant analysis (QDA). The working of both classifiers is same except that LDA assumes one covariance matrix for all classes while QDA assumes separate covariance matrix for each separate class. Both classifiers are supervised learning algorithms.

B. Support Vector Machine

Support vector machines is a classification technique which works by developing a hyper plane between the data points to be classified. The hyper plane to be developed should have the largest distance to the nearest data points. Fig. 3 shows a hyperplane and data points are classified as one can imagine that the data points are elevated in space which is done using a kernel function [25]. In its simplest form, the kernel function transforms the data into another dimension where it can be divided into different classes.

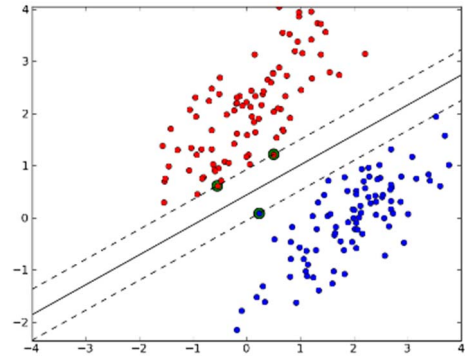


Fig. 3 Hyper plane of support vector machine

The algorithm for SVM involves:

1. Dividing the data into training and test data
2. Applying a kernel function on a training data
3. Choosing the support vectors
4. Developing a hyper plane with maximum distances from the support vectors.
5. Decision on the test data whether it is class A or class B data.
6. Checking the accuracy of classified data using a confusion matrix.

In case of two classes having good separation, it is easier to construct a hyper-plane between them. But if the classes are not separable by linear method, SVM employees the Kernel trick, which fundamentally changes the perspective of data as depicted in Fig. 4 and Fig. 5. Therefore,

SVM is the most effective method for non-linear separation. The Gaussian radial basis function (RBF) is mathematically defined as in (6) and (7):

$$K(x_i, x_j) = \Phi(x_i) \cdot \Phi(x_j) \quad (6)$$

$$K(x_i, x_j) = e^{-\|x_i - x_j\|/2\sigma^2} \quad (7)$$

Where,

$$\Phi: R^d \rightarrow H \quad (8)$$

There are different types of SVM that can be used such as Fine Gaussian SVM and Medium Gaussian SVM. Both these classifiers use Gaussian kernel function but are distinct in the values of their kernel scale which helps in scaling the kernel function for the improvement in classification.

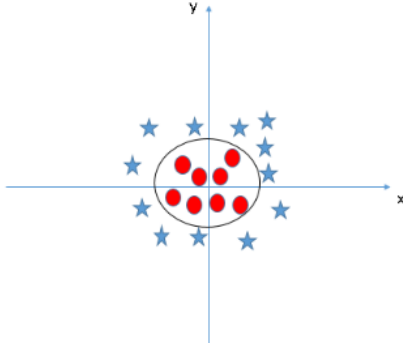


Fig. 4 Radial basis kernel

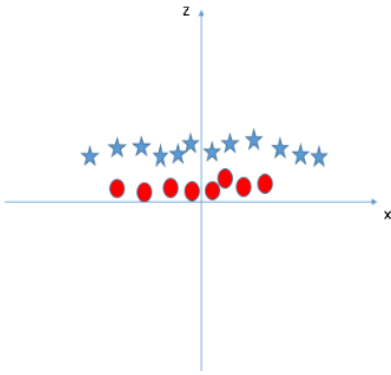


Fig. 5 Axis transformation of radial basis kernel

C. K-Nearest Neighbor classifier

It is a supervised learning classification algorithm in which the known classes are stored and new classes are classified based on the distances. These distances can be calculated using three types of distance functions which are only valid for continuous variables i.e. Euclidean, Manhattan, Minkowski [26].

$$\text{Euclidean Formula: } \sqrt{\sum_{i=1}^k (x_i - x_j)^2} \quad (9)$$

$$\text{Manhattan Formula: } \sum_{i=1}^k |x_i - x_j| \quad (10)$$

$$\text{Minkowski Formula: } [\sum_{i=1}^k (x_i - x_j)^q]^{1/q} \quad (11)$$

The two types of KNN classifiers that can be used are fine KNN and weighted KNN. Fine KNN uses $k=1$ and uses Euclidean formula to calculate the distances. Similarly,

weighted KNN uses $k=10$ and also uses Euclidean formula to calculate the distance.

VII. RESULTS AND DISCUSSIONS

EEG data was acquired for two motor imagery movements i.e. left hand and right hand using emotive headset and converted into readable data form using open vibe. MATLAB was used as a data processing tool. Three different cases were studied in which filtration, dimensionality reduction and feature extraction techniques were applied on the data and the data was classified using six classifiers i.e. LDA, QDA, fine Gaussian SVM, medium Gaussian SVM, fine KNN and weighted KNN.

TABLE I CLASSIFICATION ACCURACIES OF DIFFERENT CLASSIFIERS

Classifier	Filter	Dimension reduction	Feature extraction	Accuracy
LDA	Butterworth bandpass	---	---	45.2%
QDA	Butterworth bandpass	---	---	54.6%
Fine Gaussian SVM	Butterworth bandpass	---	---	71.2%
Medium Gaussian SVM	Butterworth bandpass	---	---	66.6%
Fine KNN	Butterworth bandpass	---	---	73.9%
Weighted KNN	Butterworth bandpass	---	---	72.1%
LDA	Butterworth bandpass	PCA	---	43.5%
QDA	Butterworth bandpass	PCA	---	56.4%
Fine Gaussian SVM	Butterworth bandpass	PCA	---	69.9%
Medium Gaussian SVM	Butterworth bandpass	PCA	---	74.1%
Fine KNN	Butterworth bandpass	PCA	---	79.6%
Weighted KNN	Butterworth bandpass	PCA	---	78.2%
LDA	Butterworth bandpass	PCA	ICA	45%
QDA	Butterworth bandpass	PCA	ICA	58.2%
Fine Gaussian SVM	Butterworth bandpass	PCA	ICA	69%
Medium Gaussian SVM	Butterworth bandpass	PCA	ICA	73.5%
Fine KNN	Butterworth bandpass	PCA	ICA	80.6%
Weighted KNN	Butterworth bandpass	PCA	ICA	78.8%

Table 1 shows accuracies for different classifiers after 5 fold cross validation. The best results were obtained by using fine KNN which uses value of K as 1. However, different classifiers were used to present a good comparison.

In the first case, Butterworth bandpass filter was applied before classification. The maximum accuracy for this case is 73.9%.

Fig. 6 shows a receiver operating characteristic (ROC) curve which is a plot for true positive rates versus false positive rates. The area under ROC curve (AUC) is 0.74 where AUC=0.5 shows a bad classification and AUC=1 shows a good classification. Similarly, the results of classification are given in table 2. There were 3072 rows of data out of which 1536 were for left hand movement and 1536 were for right hand movement.

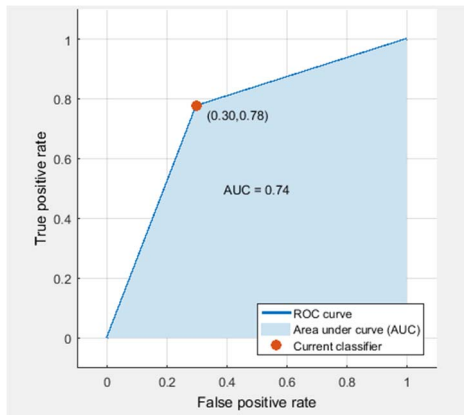


Fig. 6 ROC curve for filtered data classification

TABLE II NUMBER OF ROWS OF FILTERED CLASSIFIED DATA

Predicted class \ True class	Left hand movement (LH)	Right hand movement (RH)
Left hand movement (LH)	1192	344
Right hand movement (RH)	458	1078

In the second case, Butterworth band pass filter along with PCA technique was applied before classification. The maximum accuracy for classification achieved is 79.6%. This increase in accuracy is because of the noise reduction and complexity reduction capabilities of PCA. Fig. 7 shows the ROC curve for the classification.

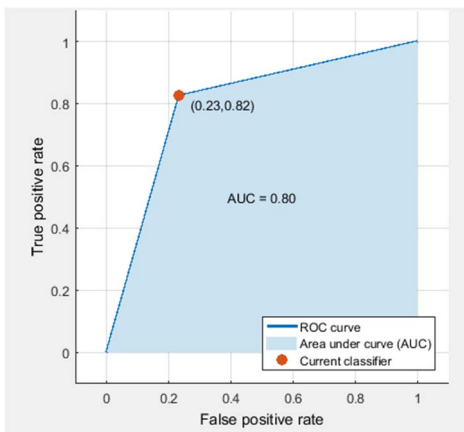


Fig. 7 ROC curve for filtered and PCA applied data classification

Here the area under the ROC curve is 0.80 which is quite acceptable value for classification. The confusion matrix for the classification is shown in Fig. 8.

TABLE III NUMBER OF ROWS OF FILTERED AND PCA APPLIED CLASSIFIED DATA

Predicted class \ True class	Left hand movement (LH)	Right hand movement (RH)
Left hand movement (LH)	1267	269
Right hand movement (RH)	359	1177

In the third case, Butterworth bandpass filter, PCA and ICA were applied before classification. The maximum accuracy for classification achieved is 80.6%. The increase in accuracy is due to the fact that ICA removes the ocular artifacts from the EEG data. Fig.9 shows the ROC curve for this case.

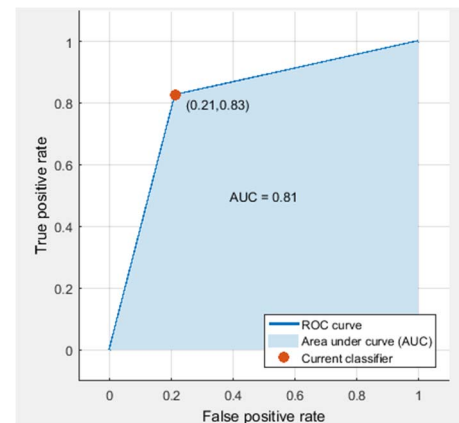


Fig. 8 ROC curve for filtered, PCA and ICA applied data classification

Here, the area under the curve is 0.81 which is the maximum for all three scenarios. The confusion matrix for this classification is shown in the Fig. 10.

TABLE IV NUMBER OF ROWS OF FILTERED, PCA AND ICA APPLIED CLASSIFIED DATA

Predicted class \ True class	Left hand movement (LH)	Right hand movement (RH)
Left hand movement (LH)	1268	268
Right hand movement (RH)	327	1209

The maximum accuracy in this case is for fine KNN which uses a value of k as 1 and is the most accurate amongst all the other classifiers.

The results of these three cases suggest that maximum classification accuracies can be achieved by applying a combination of a filter, PCA and ICA rather than applying one or two of these techniques. The maximum accuracy that was achieved in third case is 80.5% which was achieved using fine KNN as a classifier. The accuracy of same

classifier when only filter is applied is 73.9% and when filter and PCA both are applied is 79.6%. The increase in accuracy is due to the fact that PCA reduces the complexity of the data without any compromise on the information while ICA reduces the ocular artifacts from the data.

VIII. CONCLUSION

This paper presented comparison of different classification algorithms when used with and without pre-processing and feature extraction techniques. When principle component analysis (PCA) is used for dimensionality reduction, the complexity of data is reduced and thus less computational power and time is required for the classification of data. Similarly, independent component analysis (ICA) was used for feature extraction as it reduces ocular artifacts, which is the most considerable noise in BCI. The classification accuracies depicted that the best results were achieved when ocular artifacts were removed from the data. Classification was done on the data using six different classifiers. Out of all the classifiers that were used, Fine KNN gave the best accuracy because it is the most intuitive method of classification and gives minimum error rate even for large data sets.

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